Jian Feng

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EDUCATION

University of Waterloo 09/2023 – 04/2028

Bachelor of Computing and Financial Management, Honours, Co-op

Majors: Computer Science, Finance, Statistics

EXPERIENCE

Manulife 05/2026 – 08/2026

Co-op

- Triage ~10 weekly cloud/network infra tickets from global ETS teams via ServiceNow, owning resolution and stakeholder follow-through.
- Built `aks-onboarding-automation` replacing manual ServiceNow ticket exchanges; projected to cut onboarding from days to minutes.

Treasury Board of Canada Secretariat 09/2025 – 12/2025

Co-op

- Shipped a .NET MVC module that auto-normalized job titles across 5K public firms via pattern matching; cut manual data cleaning 80%.
- Migrated 30 test cases across 5 suites from Selenium to Katalon and stood up CI/CD on Azure DevOps, presenting demo to executives.
- Refactored core modules of an enterprise data platform and tuned SQL queries, cutting page load 30% and improving maintainability.

Lansi Institute of Artificial Intelligence in Medicine 04/2024 – 12/2024

Co-op (Remote)

- Built genomic ETL pipelines for ~100GB multi-omics data; per-sample fan-out via Python `multiprocessing` cut runtime 89%.
- Co-authored published T2D drug-target meta-analysis (68 studies); trained XGBoost, SMR, MR^2 to link variants to drug response.

Sanyang Technology Co., Ltd. 07/2024 – 11/2024

Intern

- Built Gantt visualizer of a flexible job-shop scheduling system, unblocking algorithm A/B runs on benchmark instances.
- Delivered CRUD APIs across multiple cloud-platform product modules, unblocking upstream and downstream system integration on schedule.

PROJECTS

ArmyOfSafeguards

- Four DeBERTa safeguard heads (toxicity, sexual, jailbreak, factuality) run in parallel via `expert_runner`, hot-swappable via env vars.
- Learned logistic meta-aggregator fuses per-expert probabilities into a calibrated verdict, beating threshold baselines on held-out HF.
- Training pipeline turns HF labels into expert-Q features and meta JSONL, so adding a new expert needs no aggregator changes.

GhostPilot

- Invisible Windows overlay streams WASAPI loopback through partial/final Azure ASR into a text LLM; watchdog auto-restarts the pipeline.
- Alt+P vision path: Qt area capture → JPEG → vision LLM, result folded back into the text-LLM prompt.
- Local RAG over `knowledge/` injects user-curated context so answers stay grounded across long sessions.

MiniKV

- Go LSM engine — CRC32C WAL, skip-list MemTable, snappy SSTables with sparse index + file-level Bloom so negative lookups skip disk.
- Serializable OCC: snapshot reads, buffered writes, commit-time read-set validation vs a bounded commit log — atomic on no conflict.
- Async leader/follower replication over HTTP NDJSON and gRPC; followers resume from local `LastSeq()` on reconnect, no event replay.

Anchor

- Splunk drift agent fingerprints a curated healthy window (volumes, templates, error rates, percentiles, top hosts) into the KV Store.
- `anchor compare` diffs a window vs anchor with LLM-narrated report + drill-in SPL; `anchor feedback` re-ranks severity from outcomes.
- `anchor history` / `anchor blind-spots` surface recurring unresolved signals as named, versioned, team-shared institutional knowledge.

ChefCopilot

- iOS cooking copilot monorepo (SwiftUI app+FastAPI ranker+ML pipeline) bound by one OpenAPI contract and shared ingredient taxonomy.
- On-device CV (YOLOv8s, 150-epoch Colab, Core ML export) scans fridge/pantry; mock fallback lets UI/backend iterate in parallel.

CabStream ETL

- Terraform-provisioned GCP stack (GCS + BigQuery + Airflow); whole pipeline tears down and rebuilds from one `apply`.
- dbt three-layer model (staging → dims/facts → marts) with `schema.yml` freshness/uniqueness tests; Airflow DAG logic unit-tested in CI.
- External-table-then-CTAS pattern keeps raw parquet cheap on GCS while marts stay query-optimised in BigQuery.

CERTIFICATES & SKILLS

Coursera: Duke Data Engineering, IBM Vector Database, UCI Database Design & Operational BI, IBM AI for Cybersecurity, Stanford Machine Learning, UCI Mathematics for ML, UCSC Bayesian Statistics, Rice Parallel-Concurrent-Distributed Programming, UA Reinforcement Learning

Skills: Python, SQL, C#, C++, Java, Git, Go, Kotlin, JavaScript, Linux, Windows, MacOS, .Net, Spring Boot, React, Node.js, Express.js, Flask, PostgreSQL, MySQL, MongoDB, Redis, AWS, GCP, Azure, Docker, Kubernetes, Terraform, GitHub Actions, Jira, Confluence, PyTorch, TensorFlow, Scikit-learn, OpenCV, GenAI tools